

Assessment

Hough circle Transform

① radius of each other

$$r = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

→ edge points: $p_1 = (5, 4)$, $p_2 = (7, 6)$

→ centres: $c_1 = (2, 4)$, $c_2 = (4, 5)$

For, $c_1 = (2, 4)$

$p_1 = (5, 4)$

$$r = \sqrt{(5-2)^2 + (4-4)^2}$$

$$r = \sqrt{3^2 + 0^2} = 3$$

using $p_2 = (7, 6)$:

$$r = \sqrt{(7-2)^2 + (6-4)^2}$$

$$= \sqrt{25+4} = \sqrt{29} = \approx 5.39.$$

So, c1 gives radd:- 3 and 5.39.

for c2 = (4,3)

using p1 = (5,4)

$$r = \sqrt{(5-4)^2 + (4-3)^2} = \sqrt{1+1} \\ = \sqrt{2} \approx 1.41$$

using p2 = (7,6):

$$r = \sqrt{(7-4)^2 + (6-3)^2}$$

$$= \sqrt{9+9} = \sqrt{18} = 4.24$$

So, c2 gives radd:- 1.41 and 4.24.

② edge point p = (8,6) centre c = (4,3)

$$r = \sqrt{(8-4)^2 + (6-3)^2} = \sqrt{16+9}$$

$$r = \sqrt{25} = r = 5$$

$$\boxed{r=5}$$

Ans:- 5

③ centre, c = (5,4), edge point p = (8,8)

$$r = \sqrt{(8-5)^2 + (8-4)^2} = \sqrt{9+16}$$

$$= \sqrt{25}$$

$$r = 5$$